SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1004	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
Student Name:	R.Nandhini	**Reg. No.:**	192524355
Date:		**Duration:**	60 minutes

Code of Conduct

I R.Nandhini, 192524355 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format:

   As a <user>, I want <feature>, so that <benefit>.
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	**Mark**
Project Vision & Stakeholder Identification (30%)	/5
Epics & User Stories (25%)	/5
Acceptance Criteria (20%)	/5
Product Backlog Prioritization & Sprint Planning (15%)	/5
Story Point Estimation, Sprint Goal & Presentation (10%)	/5
TOTAL	/25

Answer – Project 22: Intelligent Vehicle Maintenance Management System

Agile Sprint Planning & User Story Workshop

1. Project Vision and Stakeholders

1.1 Project Vision

To give fleet organizations one centralized, intelligent platform that keeps every car, bus and truck safe, available and cost-efficient, by recording service history, scheduling preventive maintenance and predicting service needs from actual vehicle usage before a breakdown happens.

Fleet organizations today often depend on paper registers, spreadsheets and memory to track services. This causes missed services, repeated breakdowns, unclear repair costs and poor visibility of which vehicles are ready for duty. The Intelligent Vehicle Maintenance Management System (IVMS) replaces these scattered methods with a single web-based system that every role in the fleet can use according to their responsibilities.

1.2 Project Objectives

- Maintain complete vehicle records and track full service history in one place.
- Schedule preventive maintenance and generate timely maintenance reminders.
- Record vehicle faults and track them until they are repaired.
- Track spare parts and repair costs, and monitor vehicle utilization.
- Generate predictive maintenance alerts and fleet maintenance reports.
- Target outcomes: fewer unexpected breakdowns, better safety, lower maintenance cost and higher fleet availability.

1.3 Scope of the Product

In scope: vehicle master data, role-based access, service history, preventive maintenance scheduling, fault reporting and tracking, reminders and predictive alerts, spare parts and cost tracking, usage logging, utilization monitoring, reports and an availability dashboard. Out of scope for this release: live GPS tracking, fuel-card integration and payroll or driver-salary management.

Agile approach: the work is delivered through Scrum with two-week sprints, a prioritized product backlog owned by the Fleet Manager, and a working, demonstrable increment at the end of every sprint.

1.4 Key Stakeholders

The following stakeholders influence or are affected by the system. Their roles and

expectations shape the epics and user stories in the next section.

Stakeholder	Role in the project	Main interest / expectation
Fleet Manager (Product Owner)	Owns the product vision and backlog; accepts completed work	Higher fleet availability, lower cost, clear reports
Maintenance Supervisor	Plans service schedules and assigns repair jobs	Accurate schedules, fault queue, workload visibility
Mechanic / Technician	Records service work, parts used and repair details	Simple data entry, full vehicle history at hand
Driver / Vehicle Operator	Reports faults and logs vehicle usage	Quick fault reporting, safe vehicle
Spare Parts / Store Manager	Maintains parts stock and issues parts	Correct stock levels, low-stock alerts
Finance / Accounts Officer	Reviews repair and maintenance expenses	Accurate cost tracking and budgeting
Operations / Dispatch Manager	Allocates vehicles to trips and routes	Know which vehicles are available or in service
Safety & Compliance Officer	Ensures vehicles meet safety rules	Proof of regular servicing, safety records
System Administrator / IT	Manages users, roles, hosting and backups	Secure, stable and maintainable system
Scrum Master and Development Team	Plan, build, test and deliver the product	Clear stories, realistic sprint scope

The Fleet Manager acts as Product Owner because this role decides what is valuable, orders the backlog and accepts each increment during the Sprint Review. The Scrum Master protects the team from interruptions and removes impediments, while the Development Team (three developers, one tester and one UI/database designer) turns stories into working software.

1.5 Assumptions, Constraints and Risks

The plan is based on the following assumptions: the team works full time on the project, the Fleet Manager is available for backlog refinement and the Sprint Review, and a test database with sample vehicles can be set up in the first week. The main

constraint is the fixed two-week sprint length, so stories that cannot be finished inside a sprint are split.

Risk	Impact	Mitigation
Vehicle data is incomplete or in different formats	Records cannot be imported; wrong service dates	Provide a data-entry template and validate every field on save
Usage readings are entered late or wrongly	Predictive alerts become unreliable	Reject lower odometer values; remind drivers daily
Requirements change during the sprint	Sprint goal is missed	Keep scope fixed; new ideas go to the product backlog
Team member absence	Velocity drops below 30 points	Keep a 1-point buffer; cross-train developers on tasks

2. Epics and User Stories

2.1 Epics

The requirements are grouped into seven epics. Each epic is a large body of work that is broken into smaller user stories that can be finished inside a sprint.

Epic	Epic name	Description
E1	Vehicle Records & Access	Maintain vehicle master data with secure role-based access
E2	Service History & Preventive Maintenance	Record services and schedule preventive maintenance
E3	Fault Management	Report, assign and track vehicle faults
E4	Reminders & Predictive Alerts	Maintenance reminders and usage-based predictive alerts
E5	Spare Parts & Cost Tracking	Track parts inventory and repair costs
E6	Vehicle Utilization	Log usage (km, hours) and monitor utilization
E7	Reports & Dashboards	Fleet maintenance reports and availability dashboard

2.2 User Stories

Every story follows the format “As a <user>, I want <feature>, so that <benefit>”. The

stories were checked against the INVEST guidelines: Independent, Negotiable, Valuable, Estimable, Small and Testable.

ID	Epic	User story
US-01	E1	As a Fleet Manager, I want to add and update vehicle records, so that all vehicle details are centralized and up to date.
US-02	E1	As a Fleet Manager, I want to search and view a vehicle profile, so that I can quickly see its details and status.
US-03	E1	As a System Administrator, I want role-based login, so that only authorized users can access or change data.
US-04	E2	As a Technician, I want to record each service done on a vehicle, so that its complete service history is tracked.
US-05	E2	As a Maintenance Supervisor, I want to schedule preventive maintenance by distance or time interval, so that vehicles are serviced before problems occur.
US-06	E3	As a Driver, I want to report a vehicle fault with a description and severity, so that the maintenance team can act quickly.
US-07	E3	As a Maintenance Supervisor, I want to assign faults to technicians and track repair status, so that no fault is left unresolved.
US-08	E4	As a Fleet Manager, I want automatic maintenance reminders for upcoming and overdue services, so that no service is missed.
US-09	E4	As a Fleet Manager, I want predictive maintenance alerts based on vehicle usage, so that I can prevent unexpected breakdowns.
US-10	E5	As a Store Manager, I want to track spare parts stock and usage, so that required parts are always available.
US-11	E5	As a Finance Officer, I want to record repair costs for each vehicle, so that maintenance spending can be controlled.
US-12	E6	As a Driver, I want to log daily kilometres and engine hours, so that vehicle usage is monitored accurately.
US-13	E6	As an Operations Manager, I want to view vehicle utilization, so that I can balance work across the fleet.
US-14	E7	As a Fleet Manager, I want fleet maintenance reports, so that I can analyse cost, downtime and service trends.
US-15	E7	As an Operations Manager, I want a fleet availability dashboard, so that I know which vehicles are ready for duty.

2.3 Non-Functional Requirements

Besides the functional stories, the following quality requirements apply to every sprint and are checked in the Definition of Done.

Quality	Requirement
Performance	Vehicle list and search results appear within 3 seconds for a fleet of up to 5,000 vehicles.
Security	Passwords are stored in hashed form; every page checks the user role before showing data.
Usability	A driver can report a fault in fewer than five clicks on a mobile browser.
Reliability	Daily database backup; system availability of at least 99% during working hours.
Maintainability	Code is reviewed, commented and covered by unit tests for all business rules.

2.4 Definition of Ready

A story is ready to enter a sprint only when it follows the Agile format, has acceptance criteria, has a story point estimate, has no unresolved dependency and can be completed by the team within one sprint.

3. Acceptance Criteria (Given – When – Then)

Acceptance criteria are written for the seven stories selected for Sprint 1. Each criterion is specific, measurable and testable, so the tester can mark it as passed or failed during the Sprint Review.

US-01 – Add / update vehicle record

- Given I am a logged-in Fleet Manager, when I enter registration number, type, make, model, year and odometer and save, then the vehicle is stored and shown in the vehicle list within 2 seconds.
- Given a registration number already exists, when I try to add it again, then the system rejects it with a “duplicate vehicle” message.
- Given a mandatory field is empty, when I save, then the system highlights the field and does not save.

US-02 – Search and view vehicle profile

- Given vehicles exist, when I search by registration number, type or status, then

matching vehicles are listed in under 3 seconds.

- Given I open a vehicle, when the profile loads, then it shows vehicle details, current status and its last service date.

US-03 – Role-based login

- Given valid credentials, when I log in, then I see only the menus allowed for my role (Admin, Manager, Technician, Driver).
- Given wrong credentials are entered 3 times, when I try again, then the account is locked for 15 minutes.
- Given I am a Driver, when I try to open the vehicle edit page, then access is denied.

US-04 – Record service history

- Given a vehicle, when the technician enters service date, odometer, work done and parts used and saves, then a service record is added to that vehicle.
- Given a service record is saved, when I view the vehicle history, then records appear in date order, latest first.
- Given the service date is in the future, when I save, then the system shows a validation error.

US-05 – Schedule preventive maintenance

- Given a vehicle, when the supervisor sets an interval (e.g. every 5,000 km or 3 months), then the next due date and due odometer are calculated automatically.
- Given a service is recorded as completed, when it is saved, then the next service due is recalculated from that service.
- Given a schedule exists, when the supervisor opens the schedule list, then upcoming and overdue services are shown separately.

US-06 – Report a vehicle fault

- Given I am a Driver, when I select a vehicle, enter fault description and severity (Low / Medium / High) and submit, then a fault ticket with a unique ID and “Open” status is created.
- Given a High severity fault is submitted, when it is saved, then the vehicle is marked “Under Inspection” and the supervisor can see it at the top of the fault list.
- Given the description is empty, when I submit, then the system will not create the ticket.

US-12 – Log vehicle usage

- Given a vehicle, when the driver enters the current odometer reading and engine

hours, then the reading is saved with date and driver name.

- Given the new odometer reading is lower than the last reading, when I save, then the system rejects it with an error.
- Given usage is logged, when I view the vehicle profile, then total distance covered is updated.

3.1 Acceptance Test Summary

Each group of acceptance criteria is converted into test cases by the tester. The table shows how many test cases cover each story and the type of testing used.

Story	Feature	Test cases	Test focus
US-03	Role-based login	6	Valid login, wrong password, 3-attempt lock-out, role access
US-01	Add / update vehicle	5	Mandatory fields, duplicate registration, save time of 2 seconds
US-02	Search / view vehicle	4	Search filters, response time of 3 seconds, profile content
US-04	Record service history	5	Record creation, latest-first order, future-date validation
US-05	Preventive schedule	6	Next-due calculation, recalculation, upcoming and overdue lists
US-06	Fault report	5	Unique ticket ID, High severity status change, empty description
US-12	Usage log	4	Reading saved, lower reading rejected, total distance update
Total		35	

4. Product Backlog Prioritization

Stories are ranked using business value, customer need and dependency. The MoSCoW method is used: M = Must have, S = Should have, C = Could have. A story that other stories depend on is ranked higher, because later work cannot start without it.

Rank	User story	Value	MoSCoW	Reason / dependency	Planned sprint
1	US-03 Role-based login	High	M	Security base; every other story depends on it	Sprint 1
2	US-01 Add / update	High	M	Vehicle master data is	Sprint 1

3	vehicle	High	M	needed by all modules	Sprint 1
4	US-04 Record service history	High	M	Core objective; depends on US-01	Sprint 1
5	US-05 Schedule preventive maintenance	High	M	Core preventive goal; depends on US-01, US-04	Sprint 1
6	US-06 Report vehicle fault	High	M	Directly improves safety and breakdown response	Sprint 1
7	US-12 Log vehicle usage	High	M	Input for reminders and predictive alerts	Sprint 1
8	US-08 Maintenance reminders	High	M	Depends on US-05 and US-12	Sprint 2
9	US-07 Assign / track faults	Medium	M	Depends on US-06	Sprint 2
10	US-10 Spare parts tracking	Medium	S	Depends on US-04	Sprint 2
11	US-11 Repair cost recording	Medium	S	Depends on US-04, US-07	Sprint 3
12	US-09 Predictive alerts	High	S	Needs usage data (US-12) and history (US-04)	Sprint 3
13	US-13 Utilization view	Medium	S	Depends on US-12	Sprint 3
14	US-14 Maintenance reports	Medium	S	Needs cost, service and fault data	Sprint 4
15	US-15 Availability dashboard	Low	C	Depends on most other modules	Sprint 4

Predictive alerts (US-09) carry high business value but are placed in Sprint 3 because they need real usage data (US-12) and service history (US-04) to work. This shows how dependencies, and not value alone, decide the order of the backlog.

4.1 Release Plan

The backlog is spread over four two-week sprints so that a usable core system is available after Sprint 1 and the intelligent features arrive in later sprints.

Sprint	Stories planned	Story points	Main outcome
Sprint 2	US-08, US-07, US-10	15	Reminders, fault tracking, parts
Sprint 3	US-11, US-09, US-13	18	Cost and predictive alerts
Sprint 4	US-14, US-15	10	Reports and dashboard

5. Sprint Backlog (Sprint 1)

Sprint length is two weeks. The team has five members: three developers, one tester and one UI/database designer. The assumed team velocity is about 30 story points per sprint. The Sprint 1 stories are divided into the development tasks below.

User story	Development tasks	Task owner
US-03 Login	Design user and role tables Build login and logout API and screen Add role-based menu and access checks Write test cases for lock-out and access	Dev 1, Tester
US-01 Vehicle records	Design vehicle table and validation rules Build add / edit vehicle form and API Duplicate registration check Unit and UI testing	Dev 2, Designer
US-02 Search / view	Build search and filter API Build vehicle list and profile page Test search response time	Dev 2
US-04 Service history	Design service record table Build service entry form and API Show history list on vehicle profile Validate dates and test	Dev 3, Tester
US-05 Preventive schedule	Define interval rules (km / months) Build next-due calculation logic Build schedule screen with upcoming / overdue view Recalculate on service completion; test	Dev 1, Dev 3
US-06 Fault report	Design fault ticket table Build fault report form for drivers Set vehicle status on High severity	Dev 3, Tester

US-12 Usage log	Test ticket creation and validation	Dev 2, Tester

5.1 Scrum Events in Sprint 1

Event	Timing	Purpose
Sprint Planning	Day 1, 2 hours	Agree the sprint goal and select the seven stories
Daily Scrum	Every day, 15 minutes	Share progress, plans and blockers
Backlog Refinement	Day 5, 1 hour	Clarify and estimate Sprint 2 stories
Sprint Review	Day 10, 1 hour	Demonstrate the increment to the Product Owner
Sprint Retrospective	Day 10, 45 minutes	Decide what to improve in the next sprint

5.2 Sprint Capacity Planning

Capacity is calculated from the working days and focus hours of each team member. A focus factor of 70% is used because part of every day goes to meetings, reviews and unplanned support. The resulting capacity supports the velocity of about 30 story points that was assumed for the sprint.

Team member	Role	Working days	Hours per day	Focus hours
Dev 1	Developer (login, schedule)	10	6	42
Dev 2	Developer (vehicle, usage)	10	6	42
Dev 3	Developer (service, fault)	10	6	42
Tester	Quality assurance	10	6	42
Designer	UI and database design	10	6	42
Total	Five members	50	30	210

6. Effort Estimation Using Story Points

Technique: Planning Poker with the Fibonacci sequence (1, 2, 3, 5, 8, 13). Each team member privately chooses a card, the cards are revealed together, and the highest and lowest estimators explain their reasoning until the team agrees. US-02 (simple search and view screen) was chosen as the reference story and given 2 points; other stories are sized relative to it, considering complexity, effort and uncertainty.

User story	Points	Justification
US-03 Role-based login	3	Standard login with roles and lock-out; well-known pattern, moderate security testing
US-01 Add / update vehicle	5	Full create / update form, validations and duplicate check across UI, API and database
US-02 Search / view vehicle	2	Reference story: read-only search and profile page, low risk
US-04 Record service history	5	New table and form, linked to vehicle, with history display and date validations
US-05 Schedule preventive maintenance	8	Most complex: interval rules, due-date calculation and recalculation logic with some uncertainty
US-06 Report vehicle fault	3	Simple form and ticket creation; added status rule for High severity
US-12 Log vehicle usage	3	Simple entry form with validation rules and total distance update
TOTAL	29	Within the assumed velocity of about 30 points

The total of 29 points is slightly below the assumed velocity of 30 points. This leaves a small buffer for defects, code review and unplanned work, so the team can commit to the sprint goal with confidence. The Fibonacci scale is used because the gap between numbers grows with size, which reflects the growing uncertainty of larger stories.

6.1 Planning Poker Round

The table shows the first-round votes of the five team members. Where the votes differed widely, the team discussed the reason and voted again until the final value was agreed.

User story	Dev 1	Dev 2	Dev 3	Test	UI	First range	Final
US-03 Role-based login	3	3	5	3	3	3 – 5	3

US-01 Add / update vehicle	5	5	5	8	3	3 – 8	5
US-05 Preventive schedule	8	13	8	8	5	5 – 13	8
US-06 Report vehicle fault	3	3	2	3	3	2 – 3	3
US-12 Log vehicle usage	3	2	3	3	5	2 – 5	3

6.2 Story Points by Epic

Epic	Stories in Sprint 1	Story points
E1 Vehicle Records & Access	US-03, US-01, US-02	10
E2 Service History & Preventive Maintenance	US-04, US-05	13
E3 Fault Management	US-06	3
E6 Vehicle Utilization	US-12	3
Total	7 stories	29

Most of the effort (23 of 29 points) is in epics E1 and E2 because vehicle records and service history are the foundation for all later features. Fault management and usage logging are small in Sprint 1 and grow in later sprints.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

“By the end of Sprint 1, authorized users can securely log in, manage vehicle records, record service history, schedule preventive maintenance, report faults and log vehicle usage in a working, tested core system.”

7.2 Expected Deliverables (Increment)

- Secure role-based login for Admin, Fleet Manager, Technician and Driver.
- Vehicle master data module with add, update, search and profile view.
- Service history module showing a full record per vehicle.
- Preventive maintenance scheduler with automatic next-due calculation.
- Fault reporting with unique ticket IDs and severity handling.
- Vehicle usage (odometer and engine hours) logging.
- Tested and demo-ready build with test cases and sprint review report.

7.3 Definition of Done

Code reviewed and merged; unit and acceptance tests passed; all acceptance criteria met; no open high-severity defects; demonstrated to the Product Owner in the Sprint Review and accepted.

Reminders, predictive alerts, parts, cost tracking, utilization views and reports are planned for Sprints 2 to 4 as per the backlog.

7.4 Sprint Review Plan

In the Sprint Review the team demonstrates each story in the order of the user journey: the administrator logs in, the Fleet Manager adds and searches a vehicle, the technician records a service, the supervisor sets a maintenance interval, a driver reports a fault and logs usage. The Product Owner accepts a story only when all its acceptance criteria are met.

7.5 Sprint 1 Output Screenshot

The Jira Sprint 1 backlog below shows the seven selected user stories with their epics, priorities and story points (total 29 points).

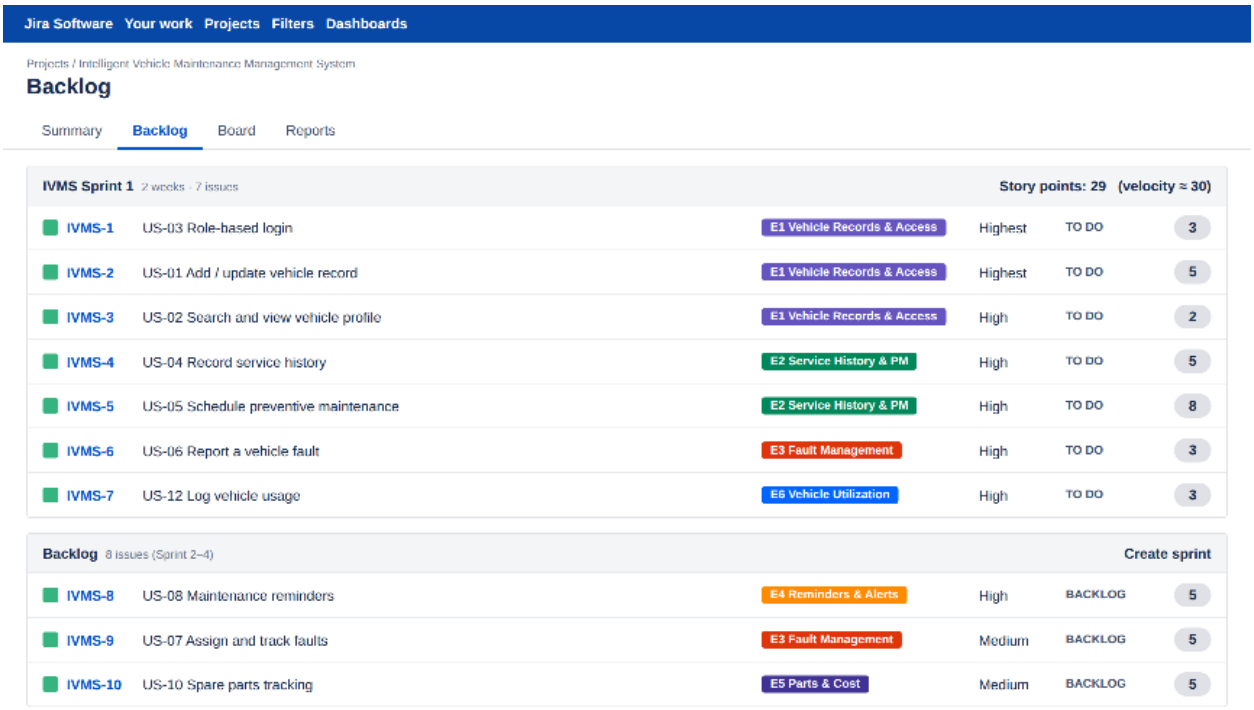


Figure 1: Jira Sprint 1 backlog (7 stories, 29 story points)